改变世界的科学巨匠

The Scientific Giant Who Changed the World



爱因斯坦因为什么获得诺贝尔奖?

英文原文

On 29 May 1919, as a total **eclipse obscured** the sun, two British **expeditions** photographed stars in its apparent **vicinity**. Arthur Eddington **observed** from Príncipe, an island off West Africa; Crommelin and Davidson from Sobral, in north-eastern Brazil. They were testing **general relativity**'s **bold** prediction that the sun's **gravity** would **deflect** starlight.

On 6 November, at a **joint** meeting of the Royal Society and the Royal Astronomical Society, the **findings** were **hailed as** confirming it. The Times **proclaimed** a "Revolution in Science"; a New York Times **headline** read "Lights All Askew in the Heavens". Almost **overnight**, Einstein, **eminent** among **physicists**, became a global **celebrity**.

Stockholm proved harder to **dazzle**. The Nobel physics committee valued **rigorous experimental confirmation**, and **committee** member Allvar Gullstrand, a Nobel-winning eye **specialist**, **deemed** the **empirical** support for relativity **insufficient**. Since no **nomination fulfilled** Nobel's **criteria**, the 1921 prize was **reserved** for a year, as the **statutes permit**.

On 9 November 1922 the Royal Swedish Academy of Sciences **conferred** it **on** Einstein, **citing** his services to theoretical physics, especially his discovery of the law of the photoelectric effect. That law, which **governs** how light **ejects electrons** from metal, helped **underpin quantum** theory. A letter the next day **clarified** that the prize **took no account of** relativity. Einstein, then **aboard** a **steamer bound for** Japan, would miss the December **ceremony**.

On 11 July 1923, in Gothenburg, Sweden, he **delivered** his Nobel **lecture** before a **congress** of Nordic scientists, **devoting** it **to** the **fundamental** ideas and problems of relativity. The theory the committee looked past reached a Nobel **podium** after all, carried there by the man the prize had **honoured**.

中文译文

1919年5月29日，一次全日食遮蔽了太阳，两支英国考察队拍摄了天空中看似位于太阳附近的恒星。阿瑟·爱丁顿观测于普林西比岛，这是一座岛，位于西非外海；克罗姆林和戴维森则在索布拉尔，位于巴西东北部。他们要检验广义相对论的一个大胆的预言，也就是太阳的引力会使经过它附近的星光偏折。

11月6日，英国皇家学会与皇家天文学会召开联席会议，观测结果被盛赞为证实了这一预言。《泰晤士报》宣告了一场“科学革命”；《纽约时报》的标题写着“天上的光全歪了”。几乎一夜之间，在物理学家中卓越的爱因斯坦成了举世皆知的名人。

斯德哥尔摩却没那么容易被眩惑。诺贝尔物理学委员会看重的是严格的实验证实，而委员会成员阿尔瓦·古尔斯特兰德，一位获得过诺贝尔奖的眼科专家，认定相对论的实证支持不足。由于没有一项提名符合诺贝尔定下的标准，1921年的奖被保留一年，这是章程所允许的。

1922年11月9日，瑞典皇家科学院把这个奖授予爱因斯坦，表彰他对理论物理学的贡献，尤其是他发现了光电效应定律。这条定律决定了光如何把电子从金属中打出，曾帮助奠定量子理论的基础。第二天的一封信澄清，此奖并未考虑相对论。爱因斯坦当时正乘一艘开往日本的轮船，将错过12月的颁奖典礼。

1923年7月11日，在瑞典哥德堡，他发表了他的诺贝尔演讲，面向北欧自然科学家大会，专门讲相对论的基本思想与问题。委员会绕开不看的那个理论，终究登上了诺贝尔的讲台，把它带上去的，正是这个奖所嘉奖的爱因斯坦。

词汇与表达

eclipse n.
日食
as a total eclipse obscured the sun,

obscured v.
遮蔽了
as a total eclipse obscured the sun,

expeditions n.
考察队
British expeditions photographed stars

vicinity n.
附近
stars in its apparent vicinity.

observed v.
观测

general relativity n.phr.
广义相对论

bold adj.
大胆的

gravity n.
引力

deflect v.
偏折
sun's gravity would deflect starlight.

joint adj.
联席

findings n.
结果
the findings were hailed as confirming

hailed as v.
盛赞为
the findings were hailed as confirming

proclaimed v.
宣告了

headline n.
标题

overnight adv.
一夜之间

eminent adj.
卓越的
Almost overnight, Einstein, eminent

physicists n.
物理学家

celebrity n.
名人

dazzle v.
眩惑

committee n.
委员会

rigorous adj.
严格的
rigorous experimental confirmation,

experimental adj.
实验

confirmation n.
证实
rigorous experimental confirmation,

specialist n.
专家

deemed v.
认定
deemed the empirical support for

empirical adj.
实证
deemed the empirical support for

词汇与表达（续）

insufficient adj.
不足
support for relativity insufficient.

nomination n.
提名

fulfilled v.
符合

criteria n.
标准
nomination fulfilled Nobel's criteria,

reserved v.
保留
the 1921 prize was reserved for a

statutes n.
章程

permit v.
允许

conferred on v.
授予

citing v.
表彰

governs v.
决定了

ejects v.
打出

electrons n.
电子

underpin v.
奠定
helped underpin quantum theory.

quantum adj.
量子

clarified v.
澄清
letter the next day clarified that the

took no account of v.phr.
并未考虑
prize took no account of relativity.

aboard prep.
乘

steamer n.
轮船

bound for adj.phr.
开往
then aboard a steamer bound for Japan,

ceremony n.
典礼

delivered v.
发表了

lecture n.
演讲

congress n.
大会

devoting to v.
专门

fundamental adj.
基本
to the fundamental ideas and problems

podium n.
讲台

honoured v.
嘉奖
by the man the prize had honoured.

文章简述

- 成名：1919年日食观测被誉为证实太阳引力会使星光偏折，这一相对论预言让爱因斯坦迅速举世闻名。
- 分歧：委员古尔斯特兰德认为相对论证据不足；1921年因无提名达标，物理学奖按章程保留一年。
- 授奖：1922年，保留的1921年度物理学奖授予爱因斯坦，表彰理论物理贡献，尤其是发现光电效应定律。
- 区别：光电效应指光使金属释放电子，这一定律帮助奠定量子理论基础；授奖决定次日的信件明确：该奖不考虑相对论。
- 演讲：爱因斯坦赴日错过颁奖典礼；1923年，他的诺贝尔演讲却专讲相对论，让被奖项绕开的理论登上讲台。